Lab: 07

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Usn No: CS23082

Aim: To apply support vector machine for classification and analyze the impact of different kernels on model accuracy

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
path="/content/drive/MyDrive/ML Dataset/pulsar_stars.csv"
df=pd.read_csv(path)
df.head()
```

	Mean of the integrated profile	Standard deviation of the integrated profile	Excess kurtosis of the integrated profile	Skewness of the integrated profile	Mean of the DM-SNR curve	Standard deviation of the DM-SNR curve	Excess kurtosis of the DM-SNR curve	Skewness of the DM-SNR curve	target_class
0	140.562500	55.683782	-0.234571	-0.699648	3.199833	19.110426	7.975532	74.242225	0
1	102.507812	58.882430	0.465318	-0.515088	1.677258	14.860146	10.576487	127.393580	0
2	103.015625	39.341649	0.323328	1.051164	3.121237	21.744669	7.735822	63.171909	0
3	136.750000	57.178449	-0.068415	-0.636238	3.642977	20.959280	6.896499	53.593661	0
4	88.726562	40.672225	0.600866	1.123492	1.178930	11.468720	14.269573	252.567306	0

Next steps: Generate code with df New interactive sheet

```
df.columns = df.columns.str.strip()
```

```
df.rename(columns={'Mean of the integrated profile': 'Mean integrated profile', 'Standard deviation of the integrated profi
```

```
print(df.columns.tolist())
```

```
['Meanoftheintegratedprofile', 'Standarddeviationoftheintegratedprofile', 'Excesskurtosisoftheintegratedprofile', 'Skewnessoftheintegratedprofile', 'MeanoftheDM-SNRcurve', 'St
```

```
print(df.columns.str.contains(' ').any())
```

```
False
```

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17898 entries, 0 to 17897
Data columns (total 9 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Meanoftheintegratedprofile    17898 non-null  float64
```

```
 1   Standarddeviationoftheintegratedprofile  17898 non-null  float64
 2   Excesskurtosisoftheintegratedprofile     17898 non-null  float64
 3   Skewnessoftheintegratedprofile           17898 non-null  float64
 4   MeanoftheDM-SNRcurve                     17898 non-null  float64
 5   StandarddeviationoftheDM-SNRcurve        17898 non-null  float64
 6   ExcesskurtosisoftheDM-SNRcurve           17898 non-null  float64
 7   SkewnessoftheDM-SNRcurve                 17898 non-null  float64
 8   target_class                             17898 non-null  int64
dtypes: float64(8), int64(1)
memory usage: 1.2 MB
None
```

df.shape

```
(17898, 9)
```

df.columns

```
Index([' Mean of the integrated profile',
       ' Standard deviation of the integrated profile',
       ' Excess kurtosis of the integrated profile',
       ' Skewness of the integrated profile', ' Mean of the DM-SNR curve',
       ' Standard deviation of the DM-SNR curve',
       ' Excess kurtosis of the DM-SNR curve', ' Skewness of the DM-SNR curve',
       'target_class'],
      dtype='object')
```

df.isnull().any().any()

```
np.False_
```

```
from sklearn.model_selection import train_test_split
X = df.drop(columns=['target_class'])
y = df['target_class']
X_train,X_test,y_train,y_test = train_test_split(X,y,train_size=0.7,test_size=0.3,random_state=42)
X_train.shape,X_test.shape
```

```
((12528, 8), (5370, 8))
```

```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

X_train

```
array([[ 0.39398903,  0.27225212, -0.09759786, ..., -0.40338108,
        -0.19481754, -0.37780071],
       [-1.32725743, -1.52127273,  0.21878252, ..., -0.44578706,
         0.23837256, -0.02857034],
       [ 0.76130595,  0.98636783, -0.3245843 , ..., -0.2004973 ,
         0.038556  , -0.29531727],
       ...,
       [ 0.19435365, -0.48635278,  0.17242235, ..., -0.75001383,
         1.92294752,  1.91898259],
       [ 0.95911539,  0.50175114, -0.53401036, ..., -0.24457614,
        -0.20616758, -0.4040674 ],
       [ 0.37725131,  0.58093396, -0.12507761, ..., -0.74159513,
         2.05677923,  2.08022728]])
```

y_train

	target_class
12223	0
1919	0
2753	0
17220	0
12110	0
...	...
11284	0
11964	0
5390	0
860	0
15795	0

12528 rows × 1 columns

dtype: int64

```
from sklearn.metrics import confusion_matrix,accuracy_score, classification_report
```

```
from sklearn.svm import SVC
svc_model = SVC(random_state=42)
svc_model.fit(X_train, y_train)
y_pred = svc_model.predict(X_test)
print('Confusion Matrix')
cm=confusion_matrix(y_pred,y_test)
print(confusion_matrix(y_pred,y_test))
```

```
Confusion Matrix
[[4863   92]
 [  21  394]]
```

```
print('Accuracy Score')
print(accuracy_score(y_pred,y_test))
```

```
Accuracy Score
0.9789571694599628
```

```
print('Confusion Matrix')
cm=confusion_matrix(y_pred,y_test)
print(confusion_matrix(y_pred,y_test))
```

```
Confusion Matrix
[[4863   92]
 [  21  394]]
```

```
import numpy as np
cm = np.array([[4863, 92],
               [21, 394]])
accuracy = np.trace(cm) / np.sum(cm)
print(f"Accuracy: {accuracy:.4f}")
```

```
Accuracy: 0.9790
```

```
class_report = classification_report(y_test, y_pred, target_names=['0', '1'])
print("Classification Report:")
print(class_report)
```

```
Classification Report:
              precision    recall  f1-score   support

           0       0.98      1.00      0.99      4884
```

```
           1       0.95      0.81      0.87       486

    accuracy                           0.98      5370
   macro avg       0.97      0.90      0.93      5370
weighted avg       0.98      0.98      0.98      5370
```

```
linear_svc=SVC(kernel='linear')
linear_svc.fit(X_train,y_train)
y_pred=linear_svc.predict(X_test)
print('model accuracy score:')
print(accuracy_score(y_pred,y_test))
```

```
model accuracy score:
0.9787709497206704
```

```
poly_svc=SVC(kernel='poly')
poly_svc.fit(X_train,y_train)
y_pred=poly_svc.predict(X_test)
```

```
sigmoid_svc=SVC(kernel='sigmoid')
sigmoid_svc.fit(X_train,y_train)
y_pred=sigmoid_svc.predict(X_test)
print(confusion_matrix(y_test,y_pred))
print(classification_report(y_test,y_pred))
print(accuracy_score(y_test,y_pred))
```

```
[[4556  328]
 [ 331  155]]
              precision    recall  f1-score   support

           0       0.93      0.93      0.93      4884
           1       0.32      0.32      0.32       486

    accuracy                           0.88      5370
   macro avg       0.63      0.63      0.63      5370
weighted avg       0.88      0.88      0.88      5370

0.8772811918063315
```

```
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report
cm=confusion_matrix(y_test,y_pred)
print(confusion_matrix(y_test,y_pred))
print(accuracy_score(y_test,y_pred))
print(classification_report(y_test,y_pred))
```

```
[[4556  328]
 [ 331  155]]
0.8772811918063315
              precision    recall  f1-score   support

           0       0.93      0.93      0.93      4884
           1       0.32      0.32      0.32       486

    accuracy                           0.88      5370
   macro avg       0.63      0.63      0.63      5370
weighted avg       0.88      0.88      0.88      5370
```

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Lab 7 Extra Datasets

Aim:To apply Support Vector Machine for Regression on Students Performance Dataset and analyze the impact of different kernels on model accuracy.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
path = "/content/drive/MyDrive/ML datasets/Student_Performance_dataset_5000.csv"
df = pd.read_csv(path)
df.head()
```

	student_id	age	gender	academic_level	study_hours	self_study_hours	online_classes_hours	social_media_hours	gaming_hours	sleep_h
0	1	18	Other	High School	7.64	1.56	2.20	3.05	2.19	
1	2	18	Other	High School	2.21	2.22	2.10	1.65	2.55	
2	3	22	Male	High School	3.45	0.00	0.29	1.34	2.08	
3	4	17	Other	High School	5.75	2.08	3.01	2.27	2.20	
4	5	19	Other	High School	6.83	1.72	3.33	2.65	0.70	

5 rows × 21 columns

```
print(df.shape)
print(df.columns)
print(df.isnull().sum())
```

```
(5000, 21)
Index(['student_id', 'age', 'gender', 'academic_level', 'study_hours',
       'self_study_hours', 'online_classes_hours', 'social_media_hours',
       'gaming_hours', 'sleep_hours', 'screen_time_hours', 'exercise_minutes',
       'caffeine_intake_mg', 'part_time_job', 'upcoming_deadline',
       'internet_quality', 'mental_health_score', 'focus_index',
       'burnout_level', 'productivity_score', 'exam_score'],
      dtype='object')
student_id            0
age                   0
gender                0
academic_level        0
study_hours           0
self_study_hours      0
online_classes_hours  0
social_media_hours    0
gaming_hours          0
sleep_hours           0
screen_time_hours     0
exercise_minutes      0
caffeine_intake_mg    0
part_time_job         0
upcoming_deadline     0
internet_quality      0
mental_health_score   0
focus_index           0
burnout_level         0
productivity_score    0
exam_score            0
dtype: int64
```

```
df["Result"] = df["exam_score"].apply(lambda x: 1 if x >= 40 else 0)
```

```
X = df.drop(columns=['exam_score', 'Result'])
y = df['Result']
```

```
X = pd.get_dummies(X, drop_first=True)
```

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)
```

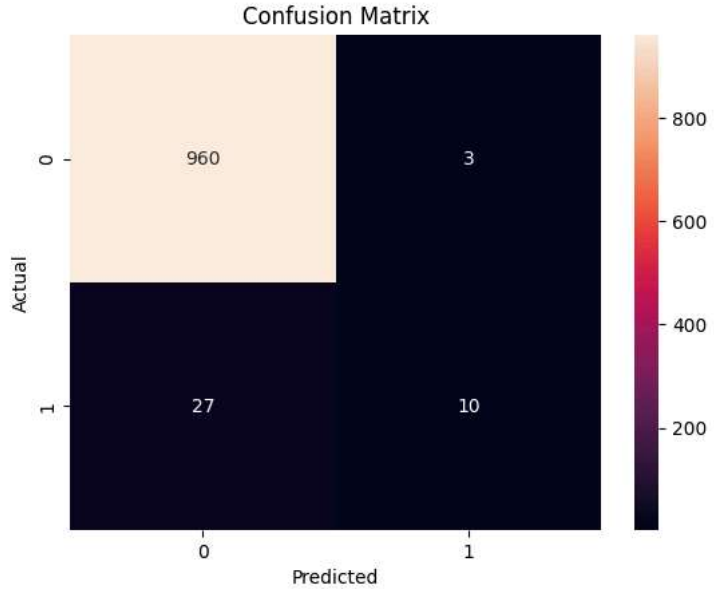
```
from sklearn.linear_model import LogisticRegression

log_model = LogisticRegression(max_iter=1000)
log_model.fit(X_train, y_train)

y_pred_log = log_model.predict(X_test)

print("Logistic Accuracy:", accuracy_score(y_test, y_pred_log))
```

```
Logistic Accuracy: 0.973
```

```
from sklearn.ensemble import RandomForestClassifier
rf_model = RandomForestClassifier()
rf_model.fit(X_train, y_train)
y_pred_rf = rf_model.predict(X_test)
print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
```

```
Random Forest Accuracy: 0.97
```

```
cm = confusion_matrix(y_test, y_pred_rf)

sns.heatmap(cm, annot=True, fmt='d')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



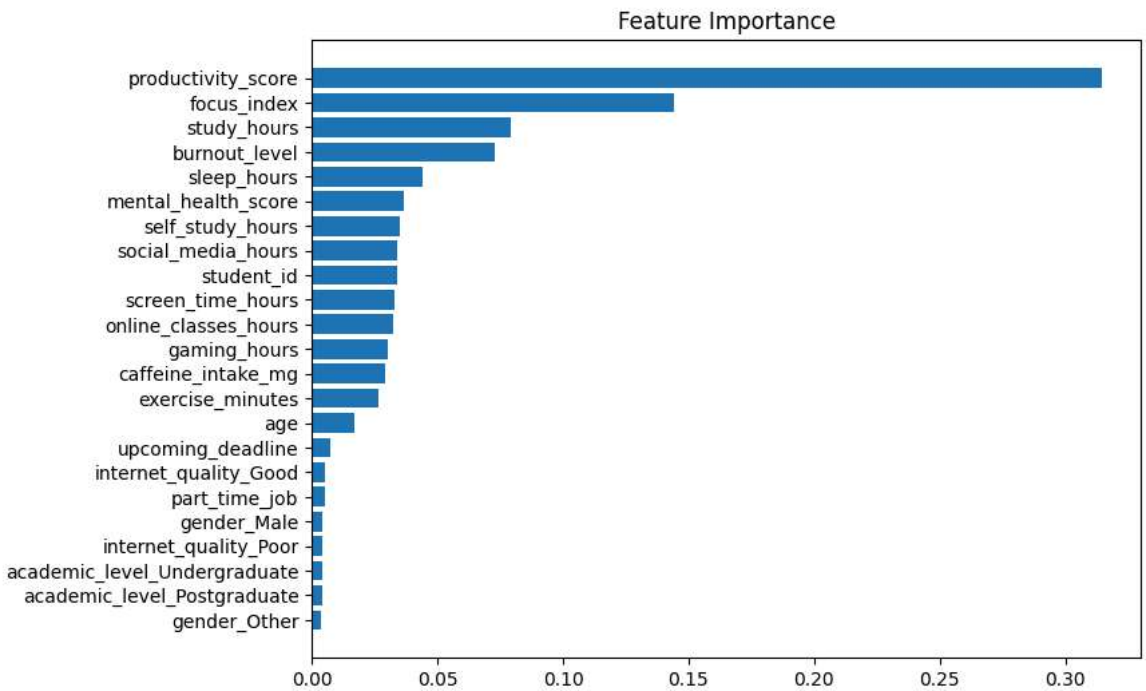

```
print(classification_report(y_test, y_pred_rf))
```

```
              precision    recall  f1-score   support

           0       0.97      1.00      0.98       963
           1       0.77      0.27      0.40        37

    accuracy                           0.97      1000
   macro avg       0.87      0.63      0.69      1000
weighted avg       0.97      0.97      0.96      1000
```

```
scores = cross_val_score(rf_model, X_scaled, y, cv=5)
print("Cross Validation Accuracy:", scores.mean())
```

```
Cross Validation Accuracy: 0.9648
```

```
importances = rf_model.feature_importances_
features = X.columns
indices = np.argsort(importances)
plt.figure(figsize=(8,6))
plt.barh(range(len(indices)), importances[indices])
plt.yticks(range(len(indices)), features[indices])
plt.title("Feature Importance")
plt.show()
```




```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)
```

```
from sklearn.svm import SVC
svc_model = SVC(kernel='rbf')
svc_model.fit(X_train, y_train)
```

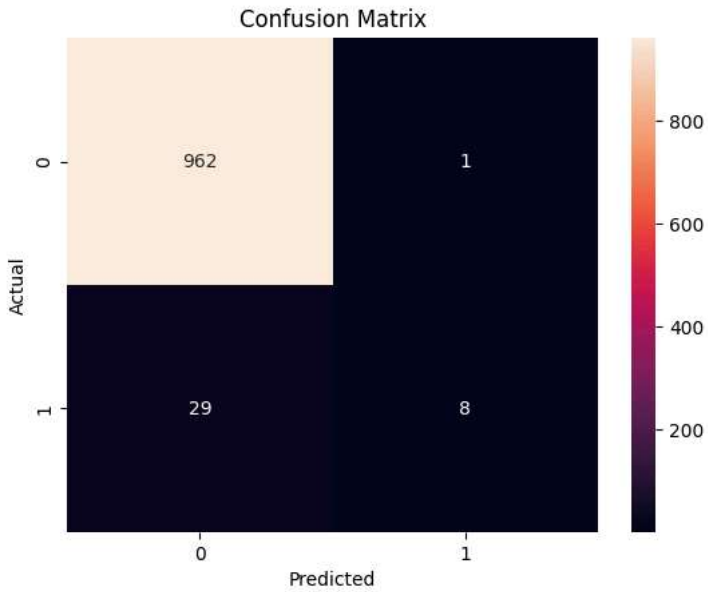
```
▾ SVC
SVC()
```

```
y_pred = svc_model.predict(X_test)
```

```
print("SVC Accuracy:", accuracy_score(y_test, y_pred))
```

```
SVC Accuracy: 0.97
```

```
cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, fmt='d')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```




```
print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.97      1.00      0.98       963
           1       0.89      0.22      0.35        37

    accuracy                           0.97      1000
   macro avg       0.93      0.61      0.67      1000
weighted avg       0.97      0.97      0.96      1000
```